Similar questions

Five new questions practising the same skills as the exam question above. They are not past exam questions. Marks are as in the original. Answers are at the end.

1 Simplify fully.

(a) $\sqrt{18a} \times \sqrt{2a^3}$ **[2]**

(b) $(27b^6)^{\frac{1}{3}} \times (9b^2)^{-\frac{1}{2}}$ **[2]**

(c) $5 \times 4^{3c} - 3 \times 8^{2c}$ **[4]**

2 Simplify fully.

(a) $\sqrt{20a^3} \times \sqrt{5a}$ **[2]**

(b) $(8b^9)^{\frac{1}{3}} \times (16b^6)^{-\frac{1}{2}}$ **[2]**

(c) $6 \times 25^{3c} - 2 \times 125^{2c}$ **[4]**

3 Simplify fully.

(a) $\sqrt{8a} \times \sqrt{2a^7}$ **[2]**

(b) $(216b^6)^{\frac{1}{3}} \times (36b^8)^{-\frac{1}{2}}$ **[2]**

(c) $8 \times 4^{3c} - 5 \times 8^{2c}$ **[4]**

4 Simplify fully.

(a) $\sqrt{27a} \times \sqrt{12a^3}$ **[2]**

(b) $(1000b^6)^{\frac{1}{3}} \times (25b^6)^{-\frac{1}{2}}$ **[2]**

(c) $10 \times 4^{2c} - 3 \times 16^{c}$ **[4]**

5 Simplify fully.

(a) $\sqrt{45a^5} \times \sqrt{5a}$ **[2]**

(b) $(343b^9)^{\frac{1}{3}} \times (49b^4)^{-\frac{1}{2}}$ **[2]**

(c) $9 \times 8^{2c} - 5 \times 4^{3c}$ **[4]**

Question 1

(a) $\sqrt{36a^4} = 6a^2$

(b) $3b^2 \times \dfrac{1}{3b} = b$

(c) $4^{3c} = 2^{6c}$ and $8^{2c} = 2^{6c}$, so $5 \cdot 2^{6c} - 3 \cdot 2^{6c} = 2 \cdot 2^{6c} = 2^{6c+1}$

Question 2

(a) $\sqrt{100a^4} = 10a^2$

(b) $2b^3 \times \dfrac{1}{4b^3} = \dfrac{1}{2}$

(c) $25^{3c} = 5^{6c}$ and $125^{2c} = 5^{6c}$, so $6 \cdot 5^{6c} - 2 \cdot 5^{6c} = 4 \cdot 5^{6c}$

Question 3

(a) $\sqrt{16a^8} = 4a^4$

(b) $6b^2 \times \dfrac{1}{6b^4} = \dfrac{1}{b^2}$

(c) $4^{3c} = 2^{6c}$ and $8^{2c} = 2^{6c}$, so $8 \cdot 2^{6c} - 5 \cdot 2^{6c} = 3 \cdot 2^{6c}$

Question 4

(a) $\sqrt{324a^4} = 18a^2$

(b) $10b^2 \times \dfrac{1}{5b^3} = \dfrac{2}{b}$

(c) $4^{2c} = 2^{4c}$ and $16^{c} = 2^{4c}$, so $10 \cdot 2^{4c} - 3 \cdot 2^{4c} = 7 \cdot 2^{4c}$

Question 5

(a) $\sqrt{225a^6} = 15a^3$

(b) $7b^3 \times \dfrac{1}{7b^2} = b$

(c) $8^{2c} = 2^{6c}$ and $4^{3c} = 2^{6c}$, so $9 \cdot 2^{6c} - 5 \cdot 2^{6c} = 4 \cdot 2^{6c} = 2^{6c+2}$